

OrbiFarm®

SERIES A · 2026

The Operating System for *indoor* agriculture.

A patented TCEA platform delivering 120x yield, -95% CAPEX, and 85%+ licensing margins — validated by Fraunhofer, protected worldwide.

STAGE

Series A

INSTRUMENT

Equity · incl. tokenize.it

HQ

Ludwigsfelde, DE

CATEGORY

AgTech · Deep Tech



OrbiFarm
Loop Your Food

THE PROBLEM

Agriculture is running out of land, water, and time.

01

Shrinking arable land

Soil degradation and climate volatility shrink viable farmland every year — while demand keeps rising.

–33%

Global topsoil lost since 1970

02

Indoor farming is broken

Vertical farms collapsed under crushing CAPEX, energy costs, and labor — InFarm, AeroFarms, Bowery.

>€50M

Typical CAPEX per facility

03

High-value crops stuck offshore

Medicinal plants, APIs, cosmetic actives — grown in unstable regions with zero supply-chain control.

€20B+

Import-dependent plant actives market

MARKET OPPORTUNITY

A **€20B+** controlled-environment market, growing double-digit.

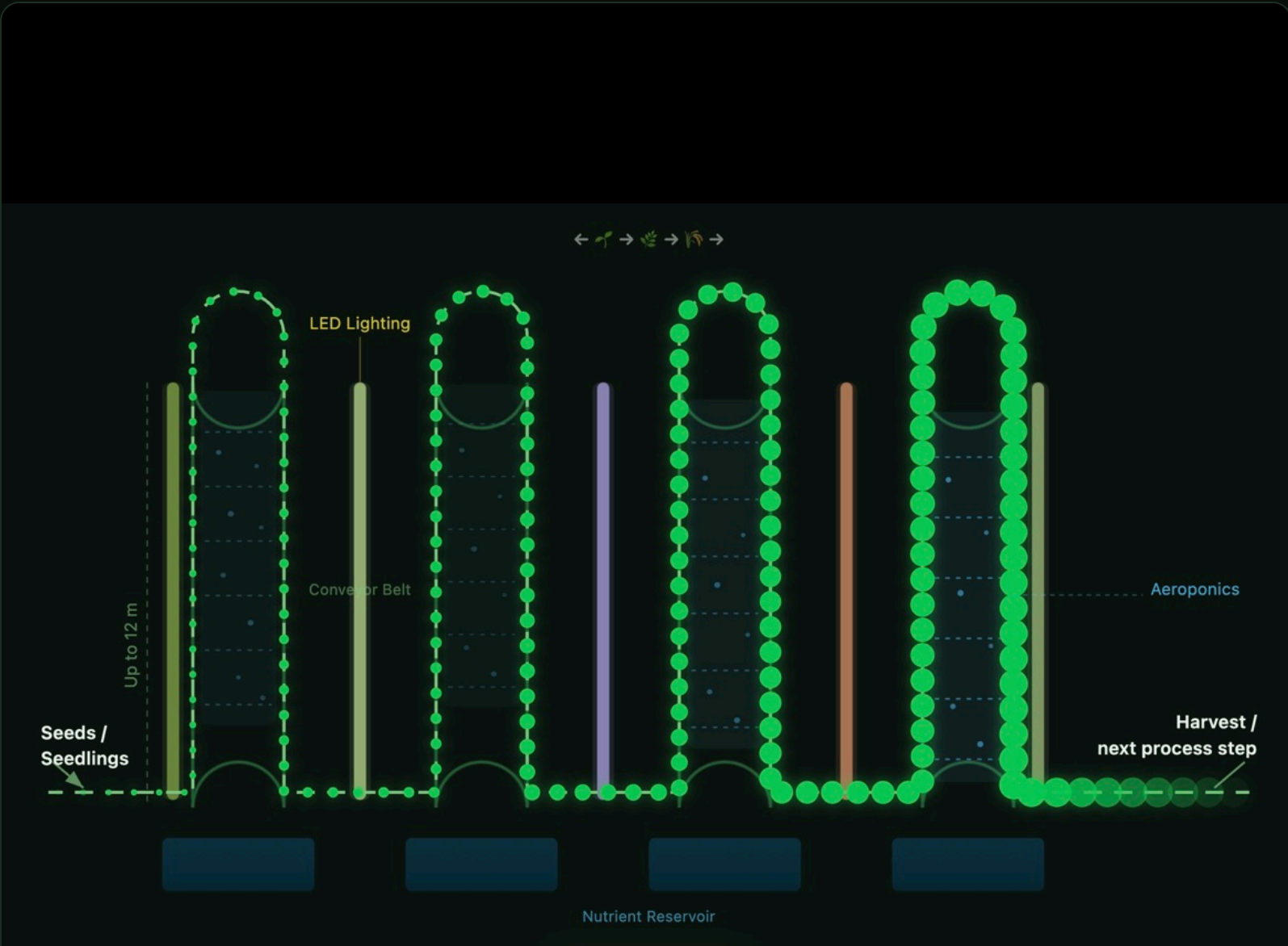
€20B TAM

Global controlled-environment agriculture (CEA) — compounding at **12–15% CAGR** through 2032. OrbiFarm sits at the intersection of three accelerating tailwinds: food security, pharma re-shoring, and sustainability mandates.

12–15% CAGR 2024–2032	3x Verticals Pharma · Food · Cosmetics
€6.5B Medicinal plants SAM by 2030	22+ Species protocols in-house
SOURCE: FORTUNE BUSINESS INSIGHTS, GRANDVIEW · 2025	

THE SOLUTION

OrbiFarm® Loop — a conveyor that **moves the plants, not the lights.**



01 — GRAVITROPISM

Plants root toward gravity

Continuously rotating plant-holders trigger a stress response that boosts biomass and active-compound density.

02 — LIGHT EFFICIENCY

One lamp feeds thousands

Fixed high-intensity lights; plants cycle past them on a conveyor. Dramatically lower energy per kg of yield.

03 — MODULAR

Container to gigafarm

Stack Loops inside a warehouse or a shipping container — identical unit economics at every scale.

Numbers that rewrite the unit economics.

120× Yield vs. outdoor per m² footprint	−95% CAPEX vs. indoor vs. conventional vertical farms	−95% Water usage closed-loop system	GMP Pharma-grade ready clean, deterministic output
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01 · ENERGY

Continuous motion = fixed lamps at peak efficiency. **70% lower kWh/kg** than static vertical farming.

02 · THROUGHPUT

Conveyor architecture enables 365-day harvesting with deterministic cycle times — minutes of downtime, not days.

03 · OUTPUT PURITY

Zero pesticides · zero herbicides · zero field contamination. GMP-traceable from seed to extract.

THE MOAT

Patents, protocols, and a three-year head start.

01 · IP

Worldwide patents granted

Core conveyor + gravitropism mechanism protected across EU, US, and key APAC jurisdictions.

02 · KNOW-HOW

22+ plant protocols

Proprietary grow recipes — each a non-replicable licensing asset.

03 · PARTNERS

Fraunhofer-validated

Joint R&D with Fraunhofer IME + Theilen Maschinenbau & Flottweg.



3

Pilot systems live

3+

Years ahead

∞

Scaling potential

TRACTION

Proven in production, not in pitch decks.

**3 pilot systems in continuous operation**

Deployed at Fraunhofer IME — live yield & energy data since 2024.

**Worldwide patents granted & secured**

Core technology IP filed in EU, US, JP, CN — freedom-to-operate confirmed.

**50× higher margins than commodity crops**

Validated on medicinal & specialty plants — priced per gram, not per tonne.

**3 strategic partners signed**

Fraunhofer IME · Theilen Maschinenbau · Flottweg

**22+ plant protocols in-house**

Ready-to-license grow recipes across pharma, food and cosmetics.

**Break-even line of sight FY28**

Clear path to profitability within 3 years on current roadmap.

BUSINESS MODEL

Three revenue streams. **One platform.**

01 · VOLUME

Hardware Sales

OrbiFarm® Loop systems — modular production units sold per site. Drives volume and market penetration.

~25%

Gross margin · upfront revenue

02 · RECURRING

Service & Maintenance

Multi-year service contracts: calibration, spare parts, process optimisation. Predictable revenue per installed unit.

~50%

Gross margin · annual

03 · HIGH MARGIN

Licensing & Royalties

Grow-recipe IP licenses, per-kilogram royalties. Scales with every deployed system — worldwide.

85%+

Gross margin · recurring IP

TARGET MARKETS

One platform, three industries.

01 · PRIORITY

Pharma & Nutraceuticals

€9.2B

Medicinal plants, APIs, cannabinoids. Highest price-per-gram, GMP-ready output — our launch vertical.

Artemisia · Salvia · Cannabis
Ginkgo · Hypericum · Echinacea

02 · SCALE

Specialty Food

€6.8B

High-value herbs, microgreens, edible flowers for HORECA & premium retail. Year-round local supply.

Basil · Cilantro · Saffron
Wasabi · Lemon balm · Chervil

03 · EXPANSION

Cosmetics & Biotech

€4.1B

Bio-active ingredients, fragrance compounds, algal & moss extracts for clean-beauty & biotech.

Rosa damascena · Centella
Calendula · Sphagnum

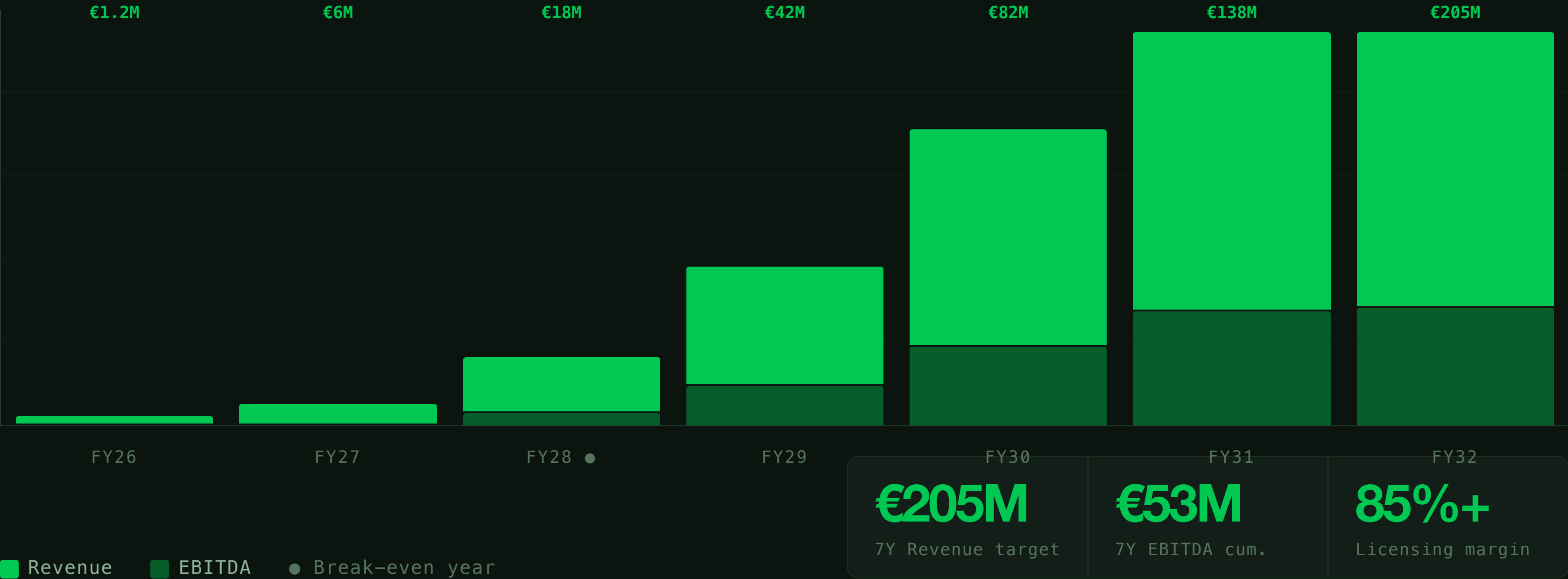
COMPETITIVE LANDSCAPE

Why we win, row by row.

METRIC	ORBIFARM® LOOP	CONVENTIONAL INDOOR / VERTICAL	OUTDOOR / FIELD
CAPEX per m²	Up to 95% lower	€8–15k / m²	Low (but no control)
Yield vs. outdoor	120×	30–50×	1×
Energy per kg	–70% vs. vertical	High (static LEDs)	Solar · free
Water usage	–95% closed loop	–70% to –80%	Very high
Pharma / GMP	• Ready	○ Retrofit required	○ Not achievable
Pesticide-free output	• Guaranteed	• Mostly	○ Rarely
Scale unit	1 Loop → gigafarm	Whole facility	Hectare blocks

Break-even FY28. €205M revenue by FY32.

\$ NDA-GATED



THE ASK

Series A • open to multiple instruments

SERIES A • SCALING ROUND → SERIES B

€2M

Series A tranche to deliver our first 10 commercial Loop installations and prepare the Series B growth round in FY27.

TOTAL FUNDING THROUGH SERIES B: €6M

INSTRUMENTS \U2014 OPEN

- Direct Equity
- SAFE
- Convertible
- tokenize.it

Structure tailored to investor preference.

USE OF FUNDS

45% Commercial roll-out
First 10 Loop installations

25% R&D • protocols
22 → 50+ grow recipes

20% Team & hires
Engineering • ops • sales

10% IP & compliance
Patent maintenance • GMP cert.

THE TEAM

Operators who've built and scaled before.

**Jan Bredack****FOUNDER · CEO**

Founder, OrbiFarm®. Serial entrepreneur in sustainable consumer & deep-tech. Managing Director, OrbiFarm GmbH.

**Dr. Jan-Gerd Frerichs****CTO**

Technology lead. Drives automation, mechatronics and platform architecture across the OrbiFarm® Loop system.

**Maja Bredack****CO-FOUNDER · COO**

Operations & organisational leadership. Scales OrbiFarm® from pilot to fleet — finance, processes, people.

**Lara Pook****HEAD OF CONSTRUCTION & DEVELOPMENT**

Leads hardware engineering and deployment. Owns Loop system design, build-out and site roll-out.

Let's grow, everywhere.

Join the Series A and secure your position in the most capital-efficient TCEA platform on the planet.

FOUNDER

Jan Bredack

WEB

orbi.farm

INVEST

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